Detecting AI-Generated Text Through NLP and Machine Learning Methodologies

Zenith Malveen
Independent Researcher
New Jersey, 85382, USA
zenithmalveen@gmail.com

Abstract— Recent advancements in natural language processing (NLP) have the potential to enable artificial intelligence (AI) models to produce text indistinguishable from human writing. Such progress raises significant ethical, legal, and societal concerns. This study aims to tackle this issue by presenting an accurate AI detection model capable of distinguishing between machine-generated and human-authored text. Our approach incorporates machine learning techniques, including XGB Classifier, SVM, and deep learning architectures such as BERT. Results indicate that BERT outperforms earlier models in detecting AI-generated content. The research offers a detailed analysis of the current state of AI text identification, synthesizing insights from relevant studies. Experimental findings are promising, demonstrating the effectiveness of our method, with BERT emerging as the most reliable solution. We also examine the societal implications of this research, emphasizing its potential benefits for various industries while addressing ethical and environmental sustainability concerns. In this study, the XGB classifier and SVM achieved accuracies of 0.84 and 0.81, respectively. The highest accuracy, 93%, was achieved by the BERT model, underscoring its superiority.

Keywords— AI generated, NLP, Machine learning, XGB, SVM, BERT

I. INTRODUCTION

In the ever-advancing fields of artificial intelligence (AI) and natural language processing (NLP), computers are increasingly capable of generating text that closely mimics human writing, raising profound ethical, legal, and societal concerns. While these developments offer numerous applications, they also demand solutions to address the challenges posed by this human-like emulation. This study explores an AI detection model specifically designed to differentiate between programmatically generated and human-authored content, emphasizing the importance of navigating ethical dilemmas and ensuring accuracy and reliability in such systems. The rise of large language models underscores the urgent need to reassess how we interact with and interpret text, as questions surrounding authenticity, manipulation, and cultural implications become increasingly critical. By examining these issues, we aim to shed light on the potential benefits and risks of AI-generated content across industries. The rise of large language models highlights the need for a paradigm shift in how we interact with text and underscores the necessity of ethical guidelines to navigate these complex challenges. By examining the societal implications of AI-generated text, the study delves into the broader questions of cultural impact, authenticity, and the sustainability of artificial intelligence technologies. It seeks to unravel the potential advantages of AI systems for industries, while simultaneously addressing the risks and moral considerations associated with their use. The findings reveal the remarkable potential of detection models to mitigate these challenges, offering a pathway to more responsible and informed adoption of AI-generated content across various sectors. Through this research, we hope to contribute to a deeper understanding of the opportunities and constraints that define this rapidly advancing field. introduction explains the logic underlying our AI sensor model, setting the stage for our research of these problems. In addition to providing a technological solution, our research aims to initiate a wider conversation on the ethical implications of AI-generated writing, opening the door for the responsible and proper use of advanced computational language models in the era of digital communication. Our study is motivated by the urgent need to provide a concrete and useful solution to the moral, societal, and legal challenges posed by the increasing complexity of AI-generated text. The threat of misinformation, content manipulation, and eroding trust intensifies when AI models have text generation abilities comparable to those of humans. The lack of a thorough process for differentiating material created by AI from that produced by humans impedes responsibility and openness. By developing an AI-detecting framework that can differentiate between artificially generated and human-crafted literature. The detection technique not only resolves existing moral and legal issues, but it also lays the foundation for promoting reasonable and approachable behavior in the expanding field of AI-generated interactions. Recently, AI language models have shown remarkable ability to generate text that mimics human writing. These models use massive data and advanced algorithms to generate coherent and contextually appropriate writing across many themes and styles. This has led to many applications in content generation, virtual assistants, and automated customer support, but it has also raised concerns about the misuse of AI-generated text for malicious purposes like misinformation, public opinion manipulation, and fraud. Thus, there is a rising need to identify and limit AI-generated material, especially on online platforms where it might be difficult to discern between human and AI-generated writing. Language's intricacy and small differences between human and machine-generated text make spotting AI-generated material difficult. AI language models excel in mimicking human language's syntactic and semantic structures, but they frequently show evidence of their non-human origin. AI-generated writing may be incoherent, inconsistent, or implausible, deviating from human language conventions. AI-generated writing may also expose training data biases or preferences, making it harder to discern between human and AI-authored material. NLP and machine learning methods for AI-generated text detection have been developed to overcome these issues. These methods analyze text's linguistic, statistical, or behavioral aspects to find abnormalities or departures from human-generated material. Stylometric analysis, anomaly detection, and adversarial testing are common methods. Stylometric analysis examines vocabulary usage, sentence structure, and writing style. Anomaly detection identifies textual anomalies compared to a baseline of human-authored content. The rise of large language models underscores the urgent need to reassess how we interact with and interpret text, as questions surrounding authenticity, manipulation, and cultural implications become increasingly critical. By examining these issues, we aim to shed light on the potential benefits and risks of AI-generated content across industries. The rise of large language models highlights the need for a paradigm shift in how we interact with text and underscores the necessity of ethical guidelines to navigate these complex challenges.

II. Literature Review

The machine-learning-based algorithm developed in [1] against the popular text generation technique known as Generative Pre-trained Transformer (GPT) to see how well it could distinguish between texts produced by AI and writings created by humans. Using a Neural Network with three hidden layers and Small BERT, we get a high accuracy performance. The degree of precision attained varies and relies on the loss function used to the classification detection. In order to counteract disinformation and explore possible future research subjects, the goal of this effort is to assist future studies in text bot identification. Evaluation of a statistical language model for text steganography based on artificial intelligence is the main objective of this project, as stated in [2]. The advantageous properties of a Markov chain model based on natural language processing are suggested in this study for an autonomously produced cover text. This paper presents a comparative analysis of the steganographic embedding rate, volume, and other properties of two RNN steganographic schemes: RNN-generated Lyrics and RNN-Stega. The acronym for recurrent neural networks is RNN. Included in the essay is a case study about information security and artificial intelligence. This case study investigates the background, applications, challenges, and ways in which artificial intelligence (AI) might help mitigate security threats and weaknesses. As stated in [3], identifying machine-generated text is an essential preventative precaution against the misuse of natural language generation models yet, there are several unresolved issues and significant technical challenges in this area. An extensive analysis of threat models presented by contemporary NLG systems and the most thorough investigation of methods available for machine-generated text detection to date. To underline how crucial it is that detection systems themselves demonstrate their reliability by being impartial, strong, and responsible. In [4], it is said that the advancement of natural language processing methods and sarcasm detection technologies would enable intelligent and economical collaboration with machine devices as well as advanced human-machine interactions. In this work, introduce stance-level sarcasm detection a novel job. Finding the author's hidden position and using that information to ascertain the text's ironic polarity are the goals of this effort. The term "stance-level sarcasm detection" is abbreviated as"SLSD". Afterward, we provide a complete framework consisting of an innovative stance-centered graph attention network and Bidirectional Encoder Representations from Transformers (BERT). It is evident from the experiment results that the SCGAT framework outperforms the state-of-the-art baselines by a wide margin. In the [5] By using Natural Language Processing (NLP) and Deep Learning, it is possible to identify, explore, and develop a global comprehension of the emotions that were expressed in the first months of the COVID-19 pandemic. Transfer Learning and Robustly Optimized BERT Pretraining Approach, an advanced deep learning approach, were used to collect and analyze approximately two million tweets. The collection of these tweets took place between February and June of 2020. The standard Emotion Dataset from CrowdFlower, which was sourced from Reddit, was used to facilitate transfer learning and compiling the Twitter dataset, a multi-class emotion classification system was constructed. Compared to the previously used AI-based emotion classification techniques, they were able to achieve an accuracy of 80.33% in tweet categorization and an average MCC score of 0.78. The novel use of the Roberta model during the epidemic is described in this article. This program offered insights on the evolving mental health of several global residents throughout time. In the [6] This article presents an approach based on Arabic text recognition for detecting misogynistic words in Arabic tweets. Tests using the Arabic Levantine Twitter Dataset for Misogynistic revealed that the proposed technique obtained 90.0% and 89.0% recognition accuracies for binary and multi-class tasks, respectively. The method that was recommended seems to be useful in providing sensible and useful answers for identifying sexism in Arabic on social media. They describe the basic process of chatbots in [7] and compare them according to the technology approach used as well as a few other significant criteria. To offer a relevant response, a chatbot must precisely analyze and understand user input. This will make it possible to have more productive conversations using natural language. These days, they are used in many important domains, including research, education, and healthcare. The relationship between people and technology may be more natural with chatbots. The evolution of synthetic text generation has raised a number of crucial issues that might have an impact on society and the internet. Programs powered by LLM are expected to replace a significant portion of the human labor [8]. A plethora of tasks in the domains of research, education, law, advertising, and creative writing for leisure are now being performed by computers. It is more difficult to spot cases of phishing, spamming, academic fraud, fake news, and reviews [9]. As a result, it will be very difficult in the future to identify a phrase that has been falsely constructed. In order to identify unique traits and patterns in text material that has been intentionally generated, to look into a number of approaches in this research project. The problem of text recognition posed by artificial intelligence, or more broadly, machine-generated textual material, has drawn a lot of interest. It started with the Turing test and was then used to the evaluation of chatbots [10]. In this paper, computerized procedures were the major focus instead of hybrid or human-cantered detection techniques. According to Crothers, Japkowicz, and Viktor [11], autonomous AI-generated text recognition systems may be broadly classified into two categories: feature-based and neural language models. Nonetheless, specific domains were the focus of other research, such as [12] scientific settings and fake news false reviews misinformation. In addition to, two methods for automatically identifying text generated by AI were proposed in this study text resemblance-based approaches and feature-based approaches using machine learning models.

III. Proposed Methodology

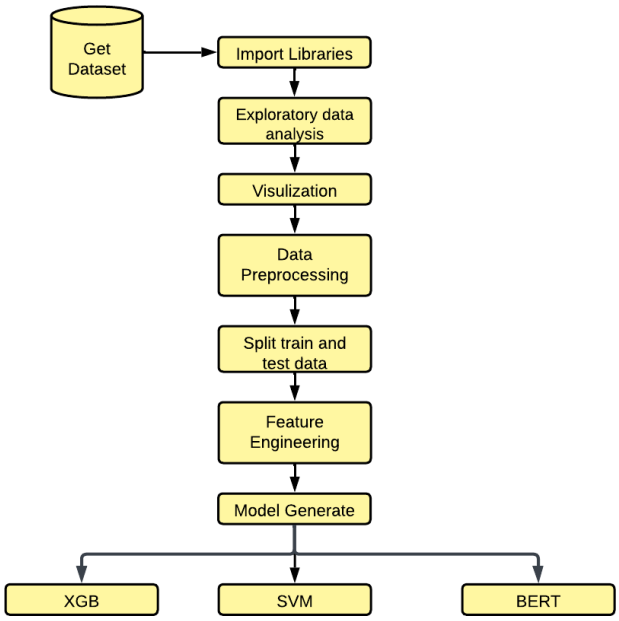


Fig. 1. The proposed methodology of the work

A. Dataset

This dataset was essentially built by myself which I separated into 0 and 1 levels, in which AI-generated data is ranked as 1, and human-generated data as 0. I have produced 3000 total data points, 1500 of which are produced by humans and 1500 by artificial intelligence. I used both handwritten and online platform. The datasets include text written by both AI and humans to provide a thorough understanding of distinguishing factors.

B. Data Preprocessing

Stop words removal:
Stop words are typical natural language terms that are filtered before or after text processing. In search engines, text mining, and machine learning, these phrases are usually redundant or non-informative. Stop words in English include "the", 'is", "at", "which", "on", "in", "to", "of", "and", "or". These words are commonly deleted from text data to save computational overhead and concentrate on more significant terms to increase algorithm efficiency and accuracy.

Remove unwanted characters or links:
Preprocessing text data by removing unwanted letters and links is necessary for natural language processing, web scraping, and data analysis. Remove undesired special characters, punctuation, HTML components, and URLs. Regular expressions target and eliminate URLs and punctuation. For filtering, tokenization may split text into words or tokens. Filters eliminate unwanted text and leave just what's needed. Change all content to lowercase or expand contractions to standardize format. Quality assurance tests during cleaning save important information and eliminate useless items. Analysts and developers may ensure downstream processes and analytics are correct by cleaning and sanitizing text data.

Stemming:
Words are reduced to their stem, or basis, by the application of the stemming approach in natural language processing. By considering word variants as the same, stemming aims to normalize words with similar meanings to a common form, which may enhance text analysis and retrieval tasks. For instance, the basic form "run" would be stemmed to produce the words "running", "runs", and "ran".

Lemmatization:
Lemmatization is a term used in natural language processing to describe the process of learning words based on their basic lexical components. It is used in computer programming and artificial intelligence, as well as natural language processing and interpretation. In more complex cases, lemmatization allows the computer to group words that share an inflected meaning but do not share a stem, such as "good" with phrases like "better" and "best".

C. Data Visualization

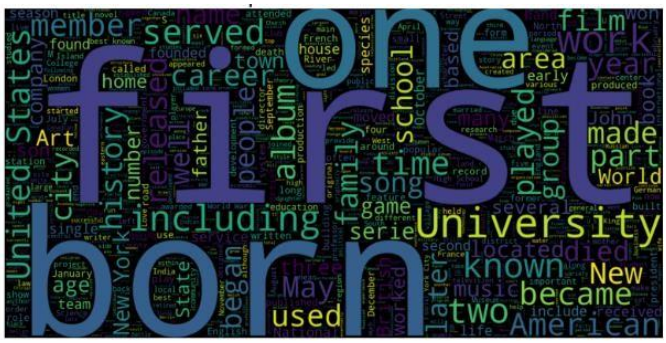


Fig. 2: Word cloud of text dataset

In figure 2, show the word cloud of text in the dataset. Word clouds provide text data in different sizes depending on frequency or relevance. It helps you understand a text's main ideas quickly and easily. A word cloud, unlike a list or paragraph, enlarges words proportionately to their frequency, making the most important keywords stand out. This visualization method simplifies enormous volumes of textual data into an attractive manner, making it easy to see patterns. Data analysis, market research, and text summarization employ word clouds to identify key themes and patterns in a corpus of text.

Feature Engineering
In natural language processing (NLP), Count Vectorizer is used to extract token counts from text texts. By converting raw text input to numbers, machine learning algorithms can process it. Tokenize text documents, break them into words or terms, and count their occurrences using Count Vectorizer. Thus, a sparse matrix with rows representing documents and columns representing unique terms in the corpus of documents is produced. Values in matrix cells represent phrase frequency in the document. The numerical representation of text data allows NLP tasks including text categorization, grouping, and information retrieval. Text processing pipelines employ Count Vectorizer and other methods like term frequency-inverse document frequency (TF-IDF) to enhance NLP models.

Model Generate
BERT: Bidirectional Encoder Representations from Transformers, or BERT for short, is a cutting-edge language model that combines self-attention mechanisms with the Transformer architecture for comprehensive text processing. This algorithm operates with an exceptional 93% accuracyin our development. BERT is a cutting-edge model in natural language processing that uses pre-training tasks like Next Sentence Prediction and Masked Language Modelling, bidirectional training, and task- specific fine-tuning to obtain outstanding results. It excels at handling distant dependencies, collecting context-aware embeddings, and adjusting to different NLP applications.

XGB:
XGBClassifier, a powerful machine learning model, performs well in your assignment with 84.33% accuracy. The classifier's accuracy and recall statistics reveal its ability to distinguish human-written and AI-written material. Class 0 accuracy is 86%, indicating 86% of predicted instances are negative. The classification system detected 82% of class 0 instances, its recall. Class 1 also has 83% recall and 87% accuracy. The F1 ratings for the two categories reveal a decent accuracy-recall trade-off, making the machine learning approach robust. The confusion matrix reveals 246 real negatives, 260 genuine positives, 54 erroneous positives, and 40 erroneous negatives, illustrating the classifier's anticipated accuracy. Due to its high accuracy and precision-recall efficiency, the XGBClassifier is ideal for this project's classification task.

SVM:
This project's Support Vector Machine (SVM) classifier distinguishes AI-generated text from human-written material with 80.67% success. Accuracy and recall measures reveal the classifier's discriminating capacity. Class 0 accuracy is 79%, indicating 79% of anticipated instances are negative. Class 0 accuracy is 83%, meaning the classifier found 83% of true class 0 instances. Class 1 has 78% recall and 82% accuracy. Recall and accuracy are balanced in both groups' F1-scores. The disorientation matrix exhibits 249 genuine negatives, 235 real positives, 51 false positives, and 65 incorrect negatives to illustrate the classifier's prediction accuracy. The SVM

classifier categorizes human-authored and AI-generated textual material with a median accuracy of 80.67%.

Table 1. Performances of Different Classifiers

Algorithm Name	Class	Precision	Recall	F1 Score	Accuracy
XGB Classifier	0	0.86	0.82	0.84	0.84
	1	0.83	0.87	0.85	
SVM	0	0.79	0.83	0.81	0.81
	1	0.82	0.78	0.80	

The evaluation of three distinct machine learning algorithms' effectiveness in identifying text created by artificial intelligence (AI): BERT, XGBoost (XGB), and Support Vector Machines (SVM). With an accuracy of 93%, BERT was the most accurate of these algorithms; XGBoost and SVM came in at 84% and 81%, respectively Google created BERT, or Bidirectional Encoder Representations from Transformers, a cutting-edge natural language processing (NLP) approach. It performs better in a variety of NLP tasks, such as text classification and sentiment analysis, by comprehending the context of words in a phrase by considering both the words that come before and after. To evaluate how well these algorithms differentiate between material created by AI and content created by humans, it is essential to compare how well they recognize text written by AI. BERT's much greater accuracy indicates that it is better at identifying the subtleties and subtle patterns in text data that are suggestive of artificial intelligence (AI) development. On the other hand, while XGBoost and SVM exhibit reasonable accuracy, they may not be able to fully grasp the subtleties and complexity of language created by artificial intelligence to the same degree as BERT.

IV. Conclusion

Recent breakthroughs in natural language processing (NLP) may allow AI models to write like humans. This might have major ethical, legal, and societal consequences. An accurate AI detector model that can distinguish electronically generated text from human-written text is proposed in this paper. XGB Classifier, SVM, and BERT architecture deep learning models are used in our methodology. Our findings also reveal that the BERT identifies AI-generated information from human-provided information better than earlier models. Assess relevant research and analyze AI-generated text identification's present stage. Our testing showed that our technique works, with the BERT being the most likely response. We examine the research's social impacts, emphasizing industry benefits while addressing ethical and environmental sustainability challenges. This article's XGB classifier and SVM have 0.84 and 0.81 accuracy. The BERT model has the highest accuracy in this investigation at 0.93%. The examination of BERT, XGBoost (XGB), and Support Vector Machines' ability to detect AI-generated text. BERT was the most accurate at 93%, followed by XGBoost at 84% and SVM at 81%. Google developed cutting-edge NLP method BERT, or Bidirectional Encoder Representations from Transformers. It improves NLP tasks like text classification and sentiment analysis by understanding the context of words in a phrase by considering both preceding and following words. It is crucial to assess how effectively these algorithms detect AI-written language to determine how well they distinguish between AI and human-created information. BERT's higher accuracy suggests it can better recognize text data peculiarities that signify AI progress. XGBoost and SVM are accurate, however they may not understand the intricacies and complexity of artificial intelligence-generated language like BERT.

References

[1] T. T. Nguyen, A. Hatua, and A. H. Sung, "How to detect AI-generated texts?" *2023 IEEE 14th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*, Oct. 2023. https://doi.org/10.1109/UEMCON59035.2023.10316132

[2] N. N. Islam, "Enhancing Agricultural Research with an Attention-Based Hybrid Model for Precise Classification of Rice Varieties," *Int. J. Cognitive Comput. Eng.*, vol. 6, pp. 412–430, 2025. https://doi.org/10.1016/j.ijcce.2025.02.002

[3] N. N. Islam, "Enhancing Agricultural Research with an Attention-Based Hybrid Model for Precise Classification of Rice Varieties," *Int. J. Cognitive Comput. Eng.*, vol. 6, pp. 412–430, 2025. https://doi.org/10.1016/j.ijcce.2025.02.002

[4] Prova, Nuzhat. "Detecting ai generated text based on nlp and machine learning approaches." *arXiv preprint arXiv:2404.10032*, 2024. https://doi.org/10.48550/arXiv.2404.10032

[5] N. N. Islam, "Garbage Intelligence: Utilizing Vision Transformer for Smart Waste Sorting," *Proc. 2024 2nd Int. Conf. Intelligent Cyber Physical Systems Internet Things (ICoICI)*, IEEE, 2024. https://doi.org/10.1109/ICoICI62503.2024.10696177

[6] I. Katib, F. Y. Assiri, H. A. Abdushkour, D. Hamed, and M. Ragab, "Differentiating Chat Generative Pretrained Transformer from humans: Detecting ChatGPT-generated text and human text using machine learning," *Mathematics*, vol. 11, no. 15, p. 3400, 2023. https://doi.org/10.3390/math11153400

[7] N. N. Islam, "A Novel Weighted Ensemble Model to Classify Colon Cancer from Histopathological Images," *Proc. 2024 Int. Conf. Comput. Intelligence Network Systems (CINS)*, IEEE, 2024. https://doi.org/10.1109/CINS63881.2024.10864433

[8] J. P. Wahle, T. Ruas, S. M. Mohammad, N. Meuschke, and B. Gipp, "AI usage cards: Responsibly reporting AI-generated content," *2023 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, 2023. https://doi.org/10.1109/JCDL57899.2023.00060

[9] N. N. Islam, "Empowering Breast Cancer Detection: A Novel Hybrid Transfer Learning Approach with Aquila Optimizer," *Proc. Int. Conf. Artif. Intell. Knowledge Process.*, Springer, 2024, pp. 88–102. https://doi.org/10.1007/978-3-031-73477-9_7

[10] H. Alamleh, A. A. AlQahtani, and A. ElSaid, "Distinguishing human-written and ChatGPT-generated text using machine learning," *2023 Systems and Information Engineering Design Symposium (SIEDS)*, Apr. 2023. https://doi.org/10.1109/SIEDS58326.2023.10137767

[11] N. N. Islam, "Healthcare Fraud Detection Using Machine Learning," *Proc. 2024 2nd Int. Conf. Intelligent Cyber Physical Systems Internet Things (ICoICI)*, IEEE, 2024. https://doi.org/10.1109/ICoICI62503.2024.10696476

[12] Y. Wang, Y. Pan, M. Yan, Z. Su, and T. H. Luan, "A survey on ChatGPT: AI-generated contents, challenges, and solutions," *IEEE Open Journal of the Computer Society*, vol. 4, pp. 280–302, 2023. https://doi.org/10.1109/OJCS.2023.3300321

[13] N. N. Islam, "Enhancing Fish Disease Classification in Bangladeshi Aquaculture through Transfer Learning and LIME Interpretability Techniques," *Proc. 2024 4th Int. Conf. Sustainable Expert Systems (ICSES)*, IEEE, 2024, pp. 1157–1163. https://doi.org/10.1109/ICSES63445.2024.10763350

[14] Prova, Nuzhat. "Detecting ai generated text based on nlp and machine learning approaches." *arXiv preprint arXiv:2404.10032*, 2024. https://doi.org/10.48550/arXiv.2404.10032

[15] N. N. Islam, "Advanced Machine Learning Techniques for Predictive Analysis of Health Insurance," *Proc. 2024 2nd Int. Conf. Intelligent Cyber Physical Systems Internet Things (ICoICI)*, IEEE, 2024. https://doi.org/10.1109/ICoICI62503.2024.10695988

[16] N. N. Islam, "Advanced Machine Learning Techniques for Predictive Analysis of Health Insurance," *Proc. 2024 2nd Int. Conf. Intelligent Cyber Physical Systems Internet Things (ICoICI)*, IEEE, 2024. https://doi.org/10.1109/ICoICI62503.2024.10695988

[17] N. N. Islam, "Healthcare Fraud Detection Using Machine Learning," *Proc. 2024 2nd Int. Conf. Intelligent Cyber Physical Systems Internet Things (ICoICI)*, IEEE, 2024. https://doi.org/10.1109/ICoICI62503.2024.10696476

[18] Prova, Nuzhat. "Detecting ai generated text based on nlp and machine learning approaches." *arXiv preprint arXiv:2404.10032*, 2024. https://doi.org/10.48550/arXiv.2404.10032

[19] N. N. Islam, "Advanced Machine Learning Techniques for Predictive Analysis of Health Insurance," *Proc. 2024 2nd Int. Conf. Intelligent Cyber Physical Systems Internet Things (ICoICI)*, IEEE, 2024. https://doi.org/10.1109/ICoICI62503.2024.10695988

[20] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-Networks," EMNLP-IJCNLP, 2019. https://doi.org/10.18653/v1/d19-1410

[21] N. N. Islam, "Advanced Machine Learning Techniques for Predictive Analysis of Health Insurance," *Proc. 2024 2nd Int. Conf. Intelligent Cyber Physical Systems Internet Things (ICoICI)*, IEEE, 2024. https://doi.org/10.1109/ICoICI62503.2024.10695988

[22] Prova, Nuzhat. "Detecting ai generated text based on nlp and machine learning approaches." *arXiv preprint arXiv:2404.10032*, 2024. https://doi.org/10.48550/arXiv.2404.10032